

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL I – Constraint-Based Problem Solving

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5 – Naturalization (BL5,BL6)	Assessment:	Assessment Tool I – Constraint Based Problem Solving
Weightage:	30% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
Student Name:	Santhosh G	Reg. No.:	192521035
Section / Batch:		Date:	24-8-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Apply backtracking techniques systematically for combinatorial problem solving.
- Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required.
- Provide proper justification, state-space representation, and complexity analysis for all solutions.
- Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions	Marks
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP-Complete.	Q1 – Q2	Q1 – 50 Q2 – 50
GRAND TOTAL:			100 Marks

SECTION A – CONSTRAINT BASED PROBLEM SOLVING

Q1. Constraint-Based Assembly Planning (NP-Hard Problem)

A manufacturing system must determine the order of assembling components under precedence constraints. Clearly define constraints such as component dependencies and assembly feasibility. Analyze how these constraints affect the ordering of operations. Develop a backtracking-based solution to generate valid assembly sequences. Apply pruning to eliminate infeasible or cyclic dependencies. Justify how your solution ensures correct and efficient assembly planning.

Q2. Constraint-Based File Replication Problem (NP-Hard Problem)

Determine file replication across servers ensuring availability and consistency. Clearly define constraints such as storage limits and replication policies. Analyze how constraints affect replication strategies. Develop a backtracking-based solution. Apply pruning to eliminate inconsistent placements. Justify correctness and discuss complexity.

Q 1. Constraint-Based Assembly Planning (NP-Hard Problem)

A manufacturing system must determine the order of assembling components under precedence constraints. Clearly define constraints such as component dependencies and assembly feasibility. Analyze how these constraints affect the ordering of operations. Develop a backtracking based solution to generate valid assembly sequences. Apply pruning to eliminate infeasible or cyclic dependencies. Justify how your solution ensures correct and efficient assembly planning.

Problem Definition:

In assembly planning a manufacturing system has a set of components/operations that must be assembled in a valid order. Some operations cannot begin until other operations have been completed.

for example:

$A \rightarrow C, B \rightarrow C, C \rightarrow D \dots$

Complexity classification:

Depending on the constraints included, the general constrained assembly-planning problem is NP-Hard. its decision form can be NP-Complete when a candidate sequence can be verified in polynomial time and the formulation captures an NP-Complete scheduling/ordering problem.

Constraints:

$$C = \{c_1, c_2, \dots, c_n\}$$

a) Precedence Constraint

$$c_i < c_j$$

b) Dependency constraint

$$\text{Prerequisite}(C_j) = \{C_i, C_k\}$$

c) Assembly feasibility constraint

$$\text{feasible}(C_i, S) = \text{true}$$

d) Resource constraint

$$r_i \leq R_{\text{max}}$$

e) Cycle constraint

$$A \rightarrow B, B \rightarrow C, C \rightarrow A$$

State-Space Representation:

$$\text{State} = (S, U)$$

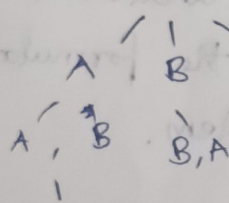
* S = Components already assembled.

* $U = C - S$ = Components remaining.

$$S = \emptyset$$

$$S = C \text{ (no more components left)}$$

$$\{ \}$$



$$A, B, C$$

$$A, B, C, D$$

Back tracking Algorithm:

Pseudo code:
ASSEMBLE-ALL (S , sequence):

if $|S| = n$:

output sequence

return

for each unassembled component c ;

if prerequisites(c) $\subseteq S$

and feasible(c, S):

add c to S

append c to sequence

ASSEMBLE-ALL (S , sequence)

remove c from sequence

remove c from S .

Pruning techniques:

Prerequisite (c_i) $\not\subseteq S$

c before A

$A \rightarrow c$

Feasible(c_i, S) = false.

* DFS-based cycle test can detect:

$A \rightarrow B \rightarrow C \rightarrow A$.

$\{c, D\}$

Memo[S] = whether a completion exists.

Example:

$A \rightarrow C$

$B \rightarrow C$

$C \rightarrow D$

Constraint

initially:

$$S = 0$$

$$S = \{A\}$$

$$\{A, B\} \notin S$$

$$S = \{A, B\}$$

$$S = \{A, B, C\}$$

$$S = \{A, B, C, D\}$$

$$\boxed{A, B, C, D}$$

$$\boxed{B, A, C, D}$$

$$A, C, B, D$$

Complexity:

Without pruning, there can be up to

$$n!$$

Worst case time complexity:

$$\boxed{O(n!)}$$

$$O(n)$$

$$\boxed{O(n2^n)}$$

depending on the constraint representation

Therefore, pruning and memoization can substantially improve practical performance, although the general problem remains exponential.

Q2. Constraint Based File Replication Problem (NP-Hard Problem).

Determine file replication across servers ensuring availability and consistency. Clearly define constraints such as storage limits and replication policies. Analyze how constraints affect replication strategies. Develop a backtracking based solution. Apply pruning to eliminate inconsistent placements. Justify correctness and discuss complexity.

Problem Definition:

* $F = \{F_1, F_2, \dots, F_m\}$: files.

* $S = \{S_1, S_2, \dots, S_k\}$: servers

$$\text{Replicas}(F_i) = \{S_a, S_b, S_c\}$$

Complexity Classification:

General constrained file-replication/placement is NP-Hard because it contains difficult combinatorial placement and resource-allocation problems as special cases.

General constrained replication = NP-Hard

NP-Complete.

Constraints:

a) Replication-factor constraint:

$$|\text{Replicas}(F_i)| = r_i$$

b) Storage Constraint:

* c_{sj} = storage capacity of server S_j .

* size_i = size of file F_i .

Then:

$$\sum_{F_i \text{ stored on } S_j} \text{size}_i \leq \text{cap}_j$$

c) Availability constraint

Suppose a file must survive at least f_i server failures:

Then:

$$r_i \geq f_i + 1.$$

d) failure - domain constraint

Replicas may need to be distributed across different racks or data centers.

$$\text{Rack}(S_a) \neq \text{Rack}(S_b).$$

e) Consistency constraint

$$\text{Version}(F_i, S_a) = \text{Version}(F_i, S_b) = \dots$$

f) Server compatibility constraint

$$\text{Allowed}(F_i, S_j) \in \{\text{true}, \text{false}\}$$

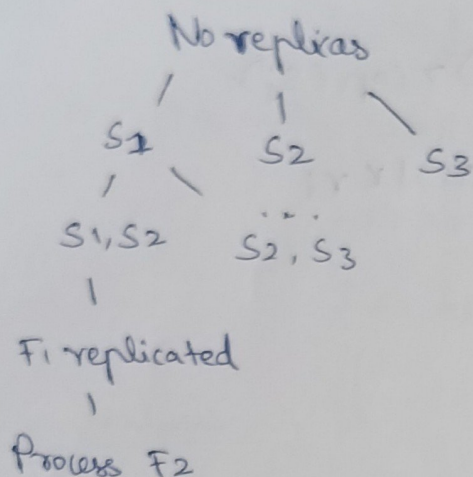
State - Space Representation:

$$\text{State} = (i, P, \epsilon)$$

Where:

- * i = current file being processed
- * P = current replica placement
- * C = remaining server capacities.

Example:



Backtracking algorithm:

REPLICATE(i):

if $i > \text{number of files}$:

return SUCCESS

(candidates = servers allowed for $F[i]$)

for each valid combination R of $r[i]$ servers:

if StorageAvailable($R, F[i]$)

and FailureProbabilityOK(R)

and CompatibilityOK(R)

and ConsistencyOK($F[i], R$):

Place $F[i]$ on servers in R

update capacities.

if REPLICATE($i+1$) = SUCCESS:

return SUCCESS

remove $F[i]$ from servers in R
restore capacities.

return FAILURE

Pruning technique:

$$\text{used}(S_j) + \text{size}(F_i) > \text{cap}(S_j)$$

$$|\text{Eligible Servers}(F_i)| < r_i$$

for example:

$$F_1 \rightarrow S_1, S_2, S_3$$

is valid if

$$S_1 \rightarrow \text{Back A}$$

$$S_2 \rightarrow \text{Back A}$$

$$S_3 \rightarrow \text{Back B}$$

$$\text{Allowed}(F_i, S_j) = \text{False}$$

Required remaining

$$\text{Free total} < \text{Required remaining}$$

Example:

Server	Capacity	Rack
S_1	100 GiB	A
S_2	100 GiB	B
S_3	100 GiB	C

$$\text{size}(F_i) = 40 \text{ GiB}$$

with replication factor:

$$r_1 = 2$$

$$\{s_1, s_2\}$$

$$\{s_1, s_3\}$$

$$\{s_2, s_3\}$$

Now suppose s_1 has only 20 GiB free

$$40 > 20$$

$$\{s_2, s_3\}$$

Complexity Analysis:

Suppose:

m = number of files:

k = number of servers

each file requires r replicas.

$$\binom{k}{r}$$

$$\left(\binom{k}{r}\right)^m$$

if r varies by file:

$$\prod_{i=1}^m \binom{k}{r_i}$$

$$k^m$$

$$\boxed{O(mk)}$$

Code of Conduct

I SANTHOSH-G-192521035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Santhosh-G

Marks Evaluation:

Question No.	Problem Understanding (25%)	Constraint Analysis (25%)	Solution Development (25%)	Application of Concepts (15%)	Justification (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided